Amirreza Sattarzadeh Khanehbargh

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Research interests: Natural Language Processing (NLP) and **Machine Learning (ML)**, centred on **language models (LMs)**: tokenization and input representations, efficient adaptation and fine-tuning, and interpretability of what models encode. Adapting NLP and ML to other domains: **BioNLP** and **protein language models (PLMs)**; **social NLP** (hate speech, multilingual and low-resource languages); financial text.

Education

M.Sc. in Computer Engineering Oct 2024 – expected Jul 2027
Boğaziçi University, Istanbul, Türkiye (English-taught)

- Supervisor: Prof. Arzucan Özgür, Text Analytics and Bioinformatics Lab (TabiLab) · GPA 3.25 / 4.0
- Thesis (in progress): *From Protein Sequences to Biological Language Units: Frequency- and Evolution-Aware Tokenization for Protein Language Models*
- Selected course projects: Turkish NLP toolkit (UD-Turkish-BOUN); demographic bias in face recognition

B.Sc. in Computer Engineering Sep 2018 – Sep 2024
Iran University of Science and Technology (IUST), Tehran, Iran

- Final project: *Financial Market Trend Analysis using Sentiment Polarity in Online Text Data* (Dr. Reza Entezari-Maleki); extended into the arXiv preprint below

High school diploma, Mathematics and Physics · GPA 19.53 / 20, ranked 1st in the school 2014 – 2018
SAMPAD (NODET: National Organization for Development of Exceptional Talents), selective gifted school, Iran

Research Experience

Graduate Research Assistant, LifeLU (ERC Consolidator Grant) Oct 2024 – present
TabiLab, Boğaziçi University · PI: Prof. Arzucan Özgür

- Study how tokenization interacts with adaptation in protein language models: amino-acid, BPE/Unigram and evolution-aware PUMA/evoBPE units with vocabularies from 800 to 51,200 tokens, under frozen-encoder, full fine-tuning and LoRA regimes, across ESM-2, ProtT5 and ProLLaMA.
- Built and ran a benchmark of **6,500+ training runs** of ESM-2 models on **14 PETA tasks** (function, localization, stability); analysed vocabulary-size effects, pooling heads, normalized perplexity, memory and compute cost. Found that mutation-aware PUMA units outperform frequency-based subwords in several settings and can match or surpass amino-acid tokens at certain vocabulary sizes.
- Tested biological knowledge fusion (predicted dynamics features added to PLM representations). Next step: interpretability of what the learned units encode.
- Weekly team reading club (one member presents a paper each week); I present every few weeks: PLM fine-tuning, tokenization methods, dynamic tokenization, infusing protein dynamics, token interpretability.

Developer and Researcher, PARI project Oct 2026 – present
Hrant Dink Foundation with Boğaziçi and Sabancı universities, Istanbul

- Open-source NLP tool for detecting discriminatory and offensive content (hate speech) in Turkish and Arabic, funded by the EU and the Friedrich Naumann Foundation; sole student researcher, working with the project's faculty, postdoctoral and PhD researchers.

Team member, ELSA Health Privacy Challenge, CAMDA 2026 at ISMB 2026 2026
Organised by EMBL, CISPA Helmholtz Center and the University of Helsinki within the EU ELSA project

- Equal-contribution first author (with B. Giledereli and B. Temiz). Benchmarked single-cell RNA-seq synthetic data generators, including an LLM-based one (Cell2Sentence), on donor-split OneK1K for biological utility versus membership-inference privacy; showed that how non-HVG genes are reconstructed is the main privacy lever (copying them: attack AUC 1.0; Poisson reconstruction: near random). Presented at ISMB 2026.

Research Assistant Jul 2023 – Sep 2024
Iran University of Science and Technology (IUST), Tehran · Supervisor: Dr. Reza Entezari-Maleki

- Financial NLP: labelled posts by the market move that follows them instead of human sentiment labels, and fine-tuned domain language models (FinBERT) on these labels for crypto trend prediction (up to +11% accuracy; 89.6% with market and temporal context in the prompt). Earlier: deep learning for algorithmic trading.

Publications and Posters

- **Sattarzadeh Khanehbargh, A.**, Suyunu, B., Dolu, Ö., and Özgür, A. (2026). *Tokenization-Aware Protein Language Modeling with Evolution-Guided Units and Biological Knowledge Fusion.* Poster, 25th European Conference on Computational Biology (**ECCB 2026**), Geneva. Presented in person.
- **Sattarzadeh Khanehbargh, A.** and Özgür, A. (2026). *From Protein Sequences to Biological Language Units: Frequency- and Evolution-Aware Tokenization for Protein Language Models.* Accepted poster, Lectures on Computational Linguistics (**LCS 2026**), Bolzano, Italy.
- Giledereli, B.*, Temiz, B.*, and **Sattarzadeh Khanehbargh, A.*** (2026). *Benchmarking Recent Single-Cell Synthetic Data Generation Methods on the OneK1K Dataset.* ELSA Health Privacy Challenge, **CAMDA 2026 at ISMB 2026**, Washington, DC (poster). Preprint, arXiv:2610.31454. *Equal contribution.
- Moradi-Kamali, H., et al., **Sattarzadeh, A.**, and Entezari-Maleki, R. (2025). *Market-Derived Financial Sentiment Analysis: Context-Aware Language Models for Crypto Forecasting.* Preprint, arXiv:2502.14897.

Industry Experience

Python Developer (Backend and Data) Jun 2022 – Nov 2023
Roshan-AI, Tehran (deep-learning company)

- Joined through a competitive backend internship. *Harf* (Persian speech recognition): built the data pipeline (crawling, cleaning, preprocessing of speech–text data), growing the training data in size and quality, which improved the speech models; Django APIs for model deployment. *Replai* (NLP comment/reply classification): Django backend.

Junior Software Engineer (freelance) Sep 2019 – Feb 2022
Bonab, Iran

- Telegram bot that crawls stock-market data and screens stocks; Django backend for a charity website (team of five).

Teaching and Mentoring

- **Teaching Assistant, Iran University of Science and Technology (IUST)** (Sep 2023 – Jul 2024), seven sections: Head TA, Data Science (Dr. H. Naderi); Software Engineering Mentor and Scrum Master, Systems Design and Analysis and Software Engineering II (Dr. M. Ashtiani, Dr. B. Minaei-Bidgoli); Advanced Programming, two sections (Dr. M. Maleki-Majd, Dr. M. Davoudabadi); Fundamentals of Programming and Computer Workshop (Dr. M. Maleki-Majd).
- **Python instructor**, online (2023 – 2024); **volunteer instructor**, educational NGO in Bonab (2023): designed and ran a free Python and computer-engineering course for students in the region.

Skills

ML / DL	PyTorch, Hugging Face Transformers, ESM-2, ProtT5, ProLLaMA, LoRA / PEFT, tokenizer design (BPE, Unigram, PUMA/evoBPE), Weights & Biases, NumPy, pandas
NLP	Text classification (offensive content, sentiment), domain LMs (FinBERT), multilingual text processing (Turkish, Persian, Arabic), speech–text data pipelines, web-scale data crawling and cleaning
Engineering	Python, Linux, remote GPU workflows (SSH), Git, SQL, Redis, Django, C, C#
Languages	Persian, Azerbaijani, Turkish (native or bilingual); English (full professional, English-taught M.Sc.); Arabic (elementary)

Honors, Tests and Service

- Iran's national graduate entrance exam (2024): ranked 764th in Computer Engineering and 398th in Information Technology; chose Boğaziçi instead. National university entrance exam (2018): 233rd of ~160,000 (Mathematics). National Mathematical Olympiad, 2nd stage (2016).
- GRE Quantitative Reasoning: 165 / 170 (2024).
- Member of ISCB and AILC (Italian Association for Computational Linguistics). Board member (since 2021) of an educational NGO in Bonab that works against educational inequality.
- Core team, Iran Innovate (Iran branch of Innovate Decentral, an international tech community), and organising team of its event Iran Innovate (Innoventiran) (2023 – 2024).

Referees

Academic (Türkiye, Iran, UK) and industry (US, Iran) references are available on request.